# A Study on Selecting Stocks Suitable for SMA Strategy Based on Fundamental and Trading Indicators

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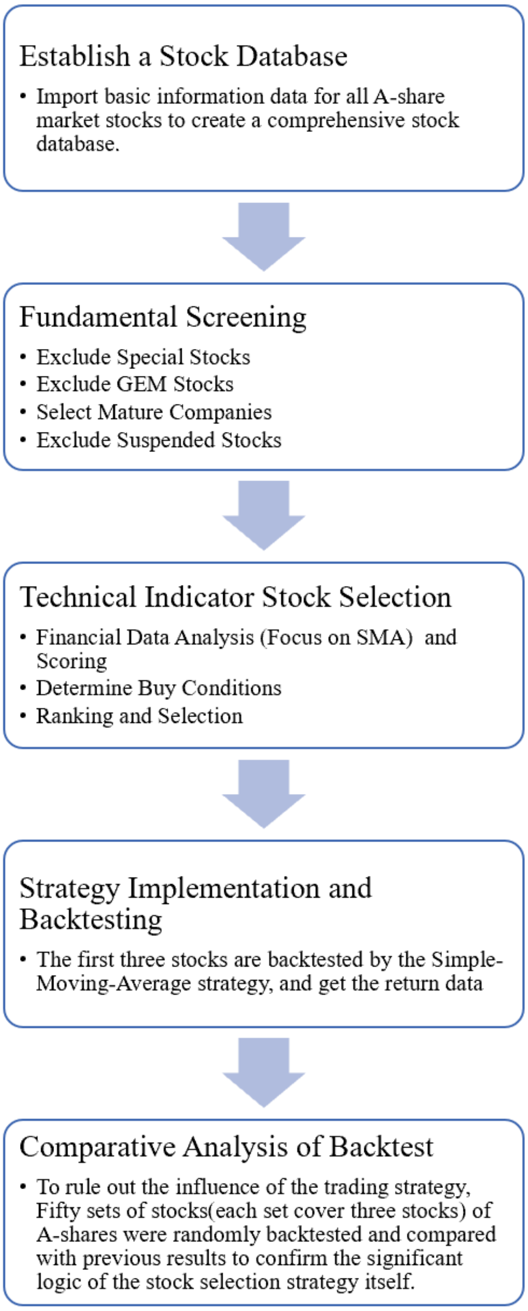
# Contents

# 1 Introduction

## 1.1 Research objective

This study aims to use a comprehensive analysis of fundamental and trading indicators to screen stocks with strong fundamentals and price trend characteristics that are suitable for the Simple Moving Average (SMA) strategy, providing investors with wiser investment choices.

## 1.2 Research approach



By establishing a stock database, implementing strict fundamental screening and technical indicator selection, backtesting and implementing strategies on selected stocks, and ultimately comparing them with a random stock selection strategy, the effectiveness of the stock selection strategy based on fundamental and trading indicators will be verified. This systematic stock selection and strategy implementation process aims to ensure that selected stocks perform well in financial stability and technical indicators, and select the most suitable stocks for SMA trading strategies to enhance investment returns.

## 1.3 Model preparation: Timeset

This research primarily selects data from 2019 to 2020 for stock selection strategies, and then conducts strategy backtesting using data from 2020 to 2021. The study chooses a recent timeline because China's overall economy is in a recovery phase in the post-pandemic era, exhibiting fluctuating and unstable conditions, which could introduce many uncontrollable factors affecting the overall strategy implementation. Therefore, the study selects a period before the pandemic when the economy was relatively stable to run the strategy, in order to more effectively assess the effectiveness of the stock selection strategy.

# 2 Model building

## 2.1 Strategic stock selection

### 2.1.1 Data collection and processing (A-share selection)

Data preprocessing aims to ensure the reliability of data and the stability of analysis results. In the preprocessing stage, this study excludes ST stocks in A-shares, ChiNext stocks, stocks listed for less than two years, and suspended stocks. Excluding specific types of stocks helps avoid unnecessary risks and uncertainties.

Here are the reasons for excluding each type of stock:

(1) Excluding ST stocks

ST (Special Treatment) stocks refer to those companies whose operational or financial conditions have become abnormal. Typically, these companies have suffered losses for two consecutive years or have poor financial conditions, thus receiving special treatment. ST stocks carry high financial risk and uncertainty, with significant market performance fluctuations, making them unsuitable for stable fundamental analysis and stock selection.

(2) Excluding ChiNext stocks

ChiNext stocks are those listed on the ChiNext market in China, primarily providing a financing platform for growth-oriented small and medium-sized enterprises and high-tech companies. ChiNext stocks usually exhibit high growth potential but also come with high risk and volatility. Their financial data may not be stable, and their performance may fluctuate significantly, which could negatively impact the stability and reliability of fundamental factor-based stock selection.

(3) Excluding stocks listed for less than two years

Stocks with a short listing period may have financial statements and operational data that do not yet fully reflect their long-term operational status and market performance, leading to significant uncertainty. This makes comprehensive fundamental analysis difficult. Therefore, excluding these stocks helps avoid analysis errors due to insufficient data.

(4) Excluding suspended stocks

Suspended stocks are those that have paused trading due to some reasons, such as major events or abnormal market fluctuations. These stocks cannot be traded during suspension.

Before fundamental stock selection, excluding ST stocks, ChiNext stocks, stocks listed for less than two years, and suspended stocks is necessary to avoid potential high risks and uncertainties. These stocks, due to unstable financial conditions, insufficient data, or trading restrictions, are not suitable as targets for a robust investment strategy. These preprocessing steps improve the accuracy and robustness of stock selection, ensuring that the selected stocks have good fundamentals and predictability.

### 2.1.2 Quantitative stock selection by fundamental factors

**Advantage**

Using fundamental factors for stock selection can avoid the blind reliance on big data, enabling comprehensive industry-wide monitoring, and striving for "stable and sustained" excess returns.

**Fundamental factor selection**

Obtain the selected stock's financial data from Tushare. The financial data is divided into three parts，including basic subjects (finance expenses/total revenue, non-interest bearing current liabilities, net debt. The above data is taken as a small value.), per share indicator factors (undistributed profit per share, earnings per share, earnings per share reserve. the above data is taken as the largest value.), value factors (turnover of account receivable, ratio of sales to cost, equity ratio. The above data is taken as a small value.)

In fundamental factor stock selection, choosing different financial indicator factors is aimed at evaluating a company's financial health and operational efficiency from multiple perspectives.

Here are the explanations for each type of factor and its specific indicators:

(1) Finance expenses/total revenue

This indicator measures the proportion of financial expenses incurred in generating revenue. A smaller value indicates that the company is more efficient in borrowing costs and financial management, with lower financial expenses, which positively impacts the company's profitability and financial health.

(2) Non-interest bearing current liabilities

Non-interest bearing current liabilities are liabilities that the company needs to repay in the short term but do not require interest payments. A smaller value indicates lower dependence on short-term liabilities, healthier liquidity, and lower financial risk.

(3) Net debt

Net debt is the balance of total debt minus cash and cash equivalents. A smaller value indicates a stronger debt repayment ability, and a more robust financial position.

(4) Undistributed profit per share

Undistributed profit per share reflects the portion of undistributed profit attributable to each share of stock. This portion of profit can be used for future reinvestment or dividends. A larger value indicates more internal accumulation, which can support future development.

(5) Earnings per share (EPS)

EPS is an important indicator of a company's profitability, reflecting the net profit attributable to each share of stock. A larger value indicates strong profitability, allowing shareholders to gain more returns.

(6) Earnings per share reserve

Earnings per share reserve is the portion of earnings reserved by the company for future expansion or contingencies. A larger value indicates good financial health, with strong capacity to cope with risks and reinvest.

(7) Turnover of accounts receivable

Turnover of accounts receivable measures the efficiency of the company in collecting its receivables. A smaller value might indicate slower collection of receivables, longer capital occupation, and poorer liquidity. Conversely, a higher turnover rate indicates better performance in credit management and capital recovery.

(8) Ratio of sales to cost

The ratio of sales to cost measures the proportion of sales cost to sales revenue. A smaller value indicates good cost control during production and sales, higher gross profit margin, and stronger profitability.

(9) Equity ratio

The equity ratio measures the proportion of a company's liabilities relative to shareholders' equity. A smaller value indicates a more stable financial structure, with lower liabilities relative to shareholders' equity, and lower financial risk.

Choosing these factors aims to comprehensively evaluate a company's financial health, profitability, operational efficiency, and financial structure. Through basic subjects and derived factors, one can understand the company's financial expenses, short-term liabilities, and net debt status. Through per share indicator factors, one can assess the company's earnings per share and undistributed profit situation. Through value factors, one can analyze the company's efficiency in collecting receivables, cost control ability, and financial structure stability. The comprehensive analysis of these factors helps to select stocks of companies that are financially healthy, highly profitable, operationally efficient, and financially stable.

**Quantitative scoring of each stock based on selected factors**

Obtain nine data points for each stock and remove outliers beyond 3 times the median. For fundamental, derivative, and value factors, classify the data using the 1st quantile method, where stocks within the 1st quantile receive a score of 1. For per-share indicator factors, classify the data using quartiles (25%, 50%, 75%), and stocks within the highest quartile receive a score of 1.

First, we perform outlier treatment (removing outliers beyond 3 times the median), aiming to reduce the impact of extreme values. Financial data often contain outliers, which may arise from data entry errors, abnormal market behavior, or other special events. These outliers can significantly distort statistical analysis results, leading to a misrepresentation of factor scores. The method of removing outliers beyond 3 times the median focuses the data around the median and excludes extreme values that are more than 3 times the median. This method can improve data reliability by eliminating extreme values, making the data more reflective of the true financial condition and market performance of the stocks, thereby avoiding the influence of a few anomalies on the overall scoring. It also reduces noise, as outliers are often noisy data; by using the outlier removal method, noise is reduced, improving the accuracy and effectiveness of factor scores.

The code is as follows:

```
def _3mad(nums):
    mad = np.median(nums)
    up = mad + 3 * mad
    down = mad - 3 * mad
    nums = np.where(nums > up, up, nums)
    nums = np.where(nums < down, down, nums)
    return nums
```

Next, we use the quantile scoring method to rate stocks. For fundamental and derivative factors as well as value factors, we classify the data according to the first quantile, dividing the data into intervals of 25%. Stocks within the first quantile range receive a score of 1. For per-share indicator factors, we categorize the data according to quartiles (25%, 50%, 75%). Stocks within the highest quartile receive a score of 1.

The code is as follows:

```
# Twenty percent quantile
cwfy_20 = np.percentile(all_dataframe[['finaexp_of_gr'][0]], 20)
wxfz_20 = np.percentile(all_dataframe[['current_exint'][0]], 20)
jzw_20 = np.percentile(all_dataframe[['netdebt'][0]], 20)

# Eighty percent of the quantile
wflr_80 = np.percentile(all_dataframe[['undist_profit_ps'][0]], 80)
shy_80 = np.percentile(all_dataframe[['q_eps'][0]], 80)
yygj_80 = np.percentile(all_dataframe[['surplus_rese_ps'][0]], 80)

# Twenty percent quantile
yszkzzl_20 = np.percentile(all_dataframe[['ar_turn'][0]], 20)
xshchb_20 = np.percentile(all_dataframe[['cogs_of_sales'][0]], 20)
chquanbl_20 = np.percentile(all_dataframe[['debt_to_eqt'][0]], 20)
```

The combination of outlier removal and the quantile method can more effectively process and analyze data in factor scoring. The outlier removal method ensures the stability and reliability of the data by eliminating extreme values, avoiding interference from anomalies. The quantile method classifies and scores the data, enabling the identification of stocks that perform best on specific indicators. This approach ensures robust data processing while enhancing the accuracy and effectiveness of factor analysis, thereby helping investors make more informed stock selection decisions.

**Results presentation**

The total score for stock rating is 9, with a maximum score of 8. Arrange the stocks in descending order based on their scores, and select stocks with scores greater than or equal to 4. Export the codes of the selected stocks to the list (Buy_list1).

### 2.1.2 Technical indicator stock selection

**Overall logic**

This section aims to construct a stock selection strategy based on different trading indicators, with the goal of selecting stocks that are most suitable for the Simple Moving Average (SMA) trading strategy. In this process, we will use the SMA trading indicator as a core factor with a higher weighting, while also combining other well-known technical indicators for a comprehensive assessment to achieve the best stock selection results.

**Technical indicator selection**

This research have selected the following eight indicators for stock analysis: Simple Moving Average (SMA), Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI), Money Flow Index (MFI), Absolute Price Oscillator (APO), Williams %R (WILLR), Average True Range (ATR), and Bollinger Bands (BOLL).

These indicators cover multiple aspects such as price trend, momentum, volatility, market sentiment, and fund flows, assisting in the evaluation of different stocks. By integrating different types of indicators, a comprehensive view of the stock's condition can be obtained, enhancing the strategy's robustness and ability to adapt to changing market conditions.

**Buy conditions for each indicator**

(1) Simple moving average (SMA)

SMA smoothens price data by calculating the average price over a certain period, making it one of the most commonly used trend-following indicators. If the stock price is above the short-term 5-day SMA and the short-term 5-day SMA is above the long-term 10-day SMA, it indicates that the stock price may be in an upward trend, meeting the buy condition.

(2) Moving average convergence divergence (MACD)

MACD uses the difference between short-term and long-term EMAs (Exponential Moving Averages) to determine the stock's trend changes, making it a momentum indicator. When the MACD line crosses above the signal line, it indicates a possible buying opportunity.

(3) Relative strength index (RSI)

RSI is a momentum indicator that measures the speed and change of stock price movements, used to determine market overbought and oversold conditions. When the RSI rises from below 30 to above 30, it indicates the end of an oversold condition and a possible price rebound, meeting the buy condition.

(4) Money flow index (MFI)

MFI combines price and volume to measure fund inflows and outflows, reflecting market fund trends. When the MFI rises from below 20 to above 20, it indicates that funds are starting to flow into the stock, potentially driving the stock price up, meeting the buy condition.

(5) Absolute price oscillator (APO)

APO determines price trend changes by calculating the difference between the current price and the average price over five days, making it a momentum indicator. When the APO rises from a negative value to a positive value, it indicates a possible trend change from a downtrend to an uptrend, meeting the buy condition.

(6) Williams (WILLR)

WILLR is an indicator that measures market overbought and oversold conditions, used to determine extreme market states. When WILLR rises from below -80 to above -80, it indicates the end of an oversold condition, potentially presenting a buying opportunity.

(7) Average true range (ATR)

ATR is used to measure market volatility, reflecting the range of price fluctuations, and is an indicator of market risk. When ATR falls below its long-term average, it indicates a decrease in market volatility, potentially presenting a favorable buying opportunity.

(8) Bollinger bands (BOLL)

Bollinger Bands reflect the price volatility range by surrounding the moving average with upper and lower bands, used to determine price volatility ranges and trends. When the stock price rebounds from the lower Bollinger Band to above the midline of the Bollinger Bands, it indicates a possible upward price reversal, meeting the buy condition.

**Quantitative scoring of each stock based on selected factors**

Each indicator has specific buy conditions. If a stock's recent data meets the buy condition of an indicator, it is assigned a score; if not, the score for that indicator is 0. The final stock selection score is obtained by adding up the scores of the eight indicators. To highlight the importance of the SMA indicator, a stock is assigned 2 points if it meets the buy condition of the SMA indicator; the rest of the indicators are assigned 1 point if they meet the buy condition, and 0 points if they do not. The total score is 8 points

**Results presentation**

After completing the scoring, we select stocks with scores greater than or equal to 5 to continue participating in the backtesting of the subsequent trading strategies. After screening, the stocks with scores greater than or equal to 5 include Wuliangye (6 points), Shanxi Fenjiu (5 points), and Zhejiang Medicine (5 points). Among them, Wuliangye has the highest score of 6 points.

| Stock Name | Stock Code | Trade Date | Score |
|---|---|---|---|
| Wuliangye | 000858.SZ | 20191231 | 6 |
| Shanxi Fen Wine | 600809.SH | 20191231 | 5 |
| Zhejiang Meida | 002677.SZ | 20191231 | 5 |

These high-scoring stocks will further participate in the backtesting of the SMA strategy to verify their investment return rates and actual trading effects.

## 2.2 SMA strategy and backtesting analysis

### 2.2.1 SMA trading strategy

Simple Moving Average (SMA) is a common technical analysis indicator used to show the trend direction of prices. In this study, we set the short-term SMA as the average value over ten days, while the long-term SMA line is set as the average value over twenty days. When the short-term SMA line crosses above the long-term SMA line, it is considered a buy signal; conversely, it is a sell signal. Based on this signal, we set the quantity for each trade at 10, meaning that we buy 10 shares each time a buy signal is generated and sell all shares when a sell signal is generated.

### 2.2.2 Backtesting

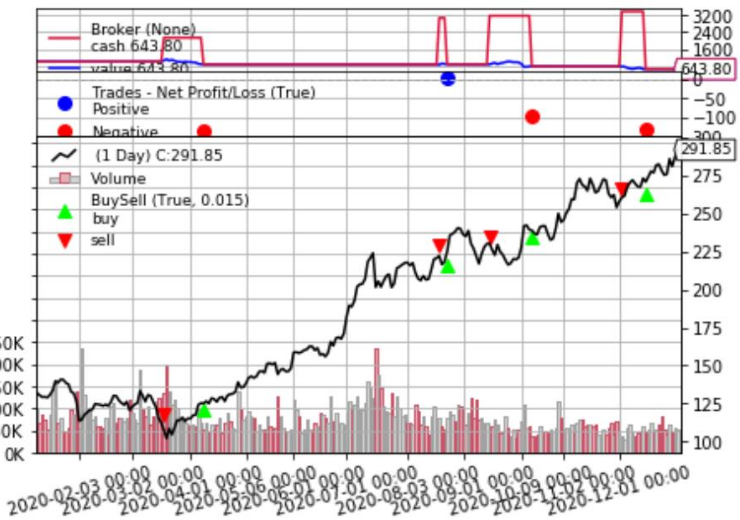
We backtested the SMA strategy using data from 2020-01-01 to 2021-01-01 and calculated the performance of this backtest.

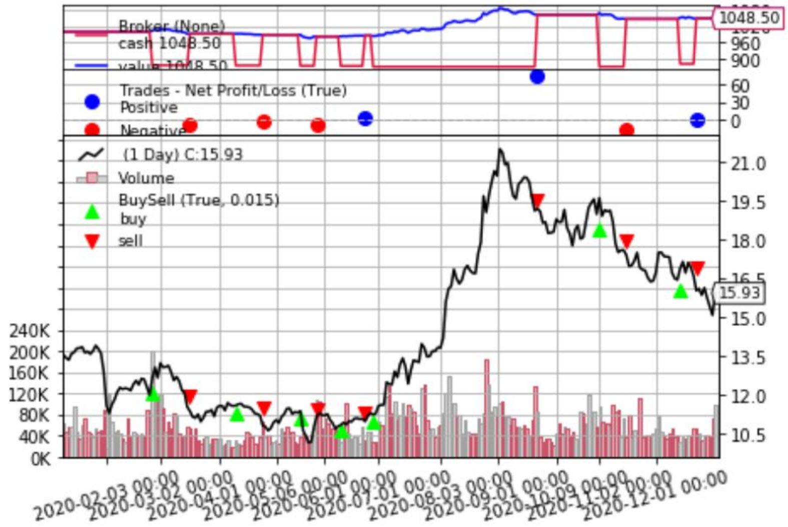
The code is as follows:

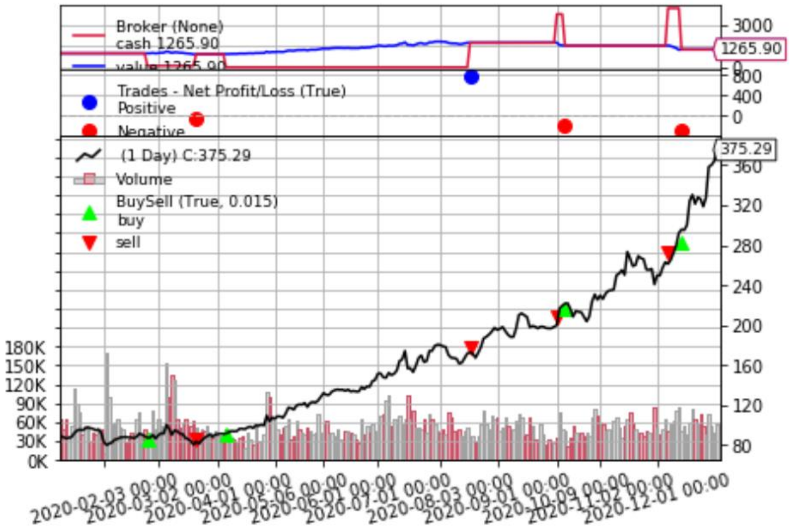
```
if __name__ == '__main__':
    print(f"Stock: {tscode}")
    cerebro = bt.Cerebro()
    #Set up initial capital
    cerebro.broker.setcash(1000.000)
    #Add stock data
    cerebro.adddata(pandasdata)
    print('Starting Portfolio Value: %.2f' % cerebro.broker.getvalue())
    #Add stratety
    cerebro.addstrategy(SMACrossoverStrategy)
    #Go through the data for that stock
    cerebro.run()
    #Draw the picture
    cerebro.plot(iplot = False)

    print('Final Portfolio Value: %.2f' % cerebro.broker.getvalue())
    print('Final Cash Value: %.2f' % cerebro.broker.getcash())
```

Strategy visualization (Wuliangye, Zhejiang Meida, Shanxi Fen Wine ):






### 2.2.3 Results presentation

| Name | Wuliangye | Shanxi Fen Wine | Zhejiang Meida |
|---|---|---|---|
| Tscode | 000858.SZ | 600809.SH | 002677.SZ |
| Industry | White wine | White wine | Household appliances |
| Annualized Return(%) | 120 | 180.6 | 20.9 |
| Maximum Drawdown | -0.1856 | -0.2654 | -0.3967 |
| Percent Profitable | 0.6467 | 0.557 | 0.4783 |
| Sharpe Ratio | 4.2 | 4.48 | 0.6 |
| Win/loss ratio | 1.83 | 1.26 | 0.92 |
| Total Return(%) | 113.9 | 170.4 | 20.1 |
| Whether successful | Yes | Yes | Yes |

**Mean annualized return rate**

$$\text{Mean} = \frac{\text{Number of profitable stocks}}{\text{Total number of selected stocks}}$$

The results show that the annualized returns of the three selected stocks are 120.0%, 180.6%, and 20.9% respectively. The average annualized return rate during the backtesting process is 107.17%.

**Ratio**

$$\text{Ratio} = \frac{\text{Number of profitable stocks}}{\text{Total number of selected stocks}}$$

In this backtesting process, we consider that achieving a total return exceeding the prevailing risk-free rate is a profitable performance, and all selected stocks have 100% profitability.

## 2.3 Comparative analysis based on random stock selection

### 2.3.1 Comparative analysis

To validate that the selected stocks are best suited for the SMA strategy, conducting a comparison with randomly selected stocks serves as a benchmark to evaluate the performance of the stock selection strategy mentioned above. By comparing with a random stock selection strategy, it can be determined whether the strategy can achieve returns in the market that outperform random investment returns.

### 2.3.2 Random stock selection

We randomly selected thirty sets, totaling 150 stocks, from the A-share stock pool to conduct SMA strategy backtesting analysis on the data for the same period and calculate the annualized return rate.

The code is as follows:

```
stock_ts_code_random = []
list = range(1,len(all_stock))
random_num = random.sample(list,len(stock_ts_code))
for num in random_num:
    stock_ts_code_random.append(all_stock[num])
```

#### 2.3.3 Results presentation

| | Mean annualized return rate(%) | Ratio(%) |
|---|---|---|
| Selected stocks | 107.17 | 100 |
| Random selected stocks | 2.53 | 38 |

Only about 38% of the total return rates of randomly selected stocks exceed the risk-free rate, with an average annualized return rate of only 2.53%, much lower than the results of the strategy stock selection. This proves the effectiveness of the stock selection strategy in this study.

# 3 Research conclusions

## 3.1 Conclusions

This study conducted an empirical research by selecting and extracting financial data and market data of some listed companies in the A-share market in China from January 1, 2019, to January 1, 2020, to construct a quantitative stock trading strategy based on fundamental factors and technical indicators. Firstly, fundamental factors such as basic items and derived factors, per share indicators, and value factors were selected. Through winsorization and scoring by quartile method, companies with strong financial health, profitability, operational efficiency, and stable financial structure were screened. Then, stocks were evaluated from different dimensions using technical indicator factors such as SMA, MACD, RSI, MFI, APO, WILLR, ATR, and BOLL to select three stocks suitable for the SMA strategy for buying. Finally, to validate this stock selection result, backtesting of the selected stocks from January 1, 2020, to January 1, 2021, using the SMA strategy was conducted, and the backtesting results were compared with those of randomly selected stocks to demonstrate the effectiveness of the stock selection strategy in this study.

## 3.2 Shortage and prospect

### 3.2.1 Shortage

**Lack of comprehensive risk assessment**

Although fundamental and technical indicators have been taken into account, the lack of a comprehensive risk assessment mechanism may lead to insufficient understanding of market and company-specific risks.

**Limitations of portfolio optimization**

While portfolio optimization has been mentioned, the specific methods and weight allocation have not been detailed, potentially resulting in unscientific and ineffective portfolio construction.

**Criteria for selection may need to be strengthened**

There may be some subjectivity in the selection of fundamental factors and technical indicators, requiring further refinement and adjustment of selection criteria to reduce randomness and enhance the universality of the strategy.

**Industry analysis may not be thorough enough**

Although industry analysis has been mentioned, specific methods and bases have not been outlined, which could lead to inaccurate or incomplete industry selection.

**Failure to consider trading costs and slippage**

Neglecting actual trading costs and slippage in backtesting may result in an overly optimistic assessment of the strategy's effectiveness.

### 3.2.2 Prospect

**Enhancing comprehensive risk assessment**

Introduce risk assessment mechanisms with more dimensions, such as market risk, industry risk, company risk, etc., to comprehensively grasp investment risks.

**Portfolio optimization methods**

Explore scientifically effective portfolio methods applied to selected multiple stocks, such as asset allocation models, Markowitz portfolio optimization models, etc., to achieve risk diversification and better returns.

**Quantification and automation of selection criteria**

Utilize data mining and machine learning technologies to improve the accuracy and universality of stock selection strategies.

**Industry analysis**

Combine industry trends and macroeconomic data to select stocks, more accurately grasp industry development trends and individual stock performance, improve stock selection accuracy and investment success rate, and achieve a more robust investment return.

**Consideration of trading costs and slippage**

Take into account actual trading costs and slippage in backtesting to more realistically evaluate the effectiveness of the strategy and potentially take corresponding optimization measures.

Through these improvement measures, we may enhance the overall effectiveness of the stock selection strategy, enabling the strategy to yield positive results in a more volatile market environment as well.

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